

Solution Requirements

Date	15 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Farmers and administrators can securely log in using username and password.</i>	High
2	Authorization levels	<i>Farmers can view recommendations; Administrators can manage datasets and system settings.</i>	High
3	External interfaces	<i>System integrates with weather data services and agricultural datasets for environmental information.</i>	High

4	Transactions processing	<i>System collects soil and weather inputs, processes them, and generates crop recommendations.</i>	High
5	Reporting	<i>Generate crop recommendation reports, yield prediction reports, and dashboard summaries.</i>	Medium
6	Business rules, etc.	<i>Crop recommendations are generated based on N, P, K values, temperature, humidity, pH, and rainfall.</i>	High
7	Compliance to laws or regulations	<i>User data must be handled according to applicable data privacy and security regulations.</i>	Medium
8	Other	<i>System provides crop optimization suggestions and resource utilization guidance.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>System should generate crop recommendations quickly.</i>	<i>Response time < 3 seconds</i>
2	Scalability	<i>System should support increasing numbers of users and agricultural records.</i>	<i>Supports 1,000+ users</i>
3	Security & Data Privacy	<i>Protect user and agricultural data through secure storage and access control.</i>	<i>Encrypted data storage and secure login</i>
4	Reliability & Availability	<i>System should remain operational with minimal downtime.</i>	<i>99% uptime per month</i>
5	Usability & Accessibility	<i>Interface should be simple and user-friendly for farmers and agricultural experts.</i>	<i>Easy navigation and responsive design</i>
6	Other	<i>Maintainability and portability across different devices and platforms.</i>	Supports web and mobile access